What Positional Frequencies Do Encoder-Decoder MT Models Need? An Empirical Study via Adaptive RoPE

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***Abstract*—Rotary Position Embedding (RoPE) [1] has become the de facto positional encoding in modern neural machine translation and large language models, yet its assumption that all frequency components contribute equally to position representation has not been empirically validated. We introduce Adaptive RoPE, a minimal extension that augments RoPE with learnable per-head frequency gates and phase offsets. The gates are initialised to one (recovering standard RoPE) and trained by gradient descent, allowing us to measure which frequency components the model actually uses. Experiments on WMT14 English-German with an encoder-decoder Transformer show that trained gates consistently converge to values below one (mean $\approx$ 0.90), indicating that models prefer low-frequency positional structure over high-frequency components. Adaptive RoPE matches standard RoPE in translation quality (BLEU 21.43 vs. 21.45) while achieving better model calibration (validation loss 0.753 vs. 0.756) at essentially zero computational overhead (+96 parameters). These results suggest that RoPE's uniform frequency weighting is a suboptimal inductive bias for machine translation, and that the frequency preference is a learnable, structured property that varies systematically across attention heads.**

***Index Terms*—rotary position embedding, machine translation, positional encoding, frequency analysis, encoder-decoder transformer**

I. Introduction

Position-aware attention is central to sequence-to-sequence models. Sinusoidal absolute encodings [2] were the original solution, but relative and rotary approaches have largely superseded them in modern architectures [1], [3], [4].

Rotary Position Embedding (RoPE) [1] encodes position by rotating query and key vectors in two-dimensional frequency planes. Each dimension pair $(2i, 2i{+}1)$ is rotated by an angle $\theta_p^{(i)} = p \cdot \omega_i$, where $\omega_i = 10000^{-2i/d}$ is a fixed frequency. Critically, *all* frequency components receive weight one — no component is treated as more or less important than another. RoPE is now the positional encoding of choice in state-of-the-art language models [5], [6].

A recent study on decoder-only language models [7] showed empirically that high-frequency RoPE components contribute less to attention than low-frequency ones and can be truncated without significant quality loss. However, it is unknown whether the same frequency preference appears in *encoder-decoder* machine translation models, which differ fundamentally in that positional information must serve both a bidirectional encoder and a causal decoder with cross-attention.

We address this gap by designing Adaptive RoPE: a learnable probe that adds trainable per-head gates $g_q, g_k \in \mathbb{R}^{H\times F}$ and phase offsets $\phi_q, \phi_k \in \mathbb{R}^{H\times F}$ to the standard RoPE rotation angles. When gates are fixed at one and phases at zero, Adaptive RoPE is identical to RoPE. Trained by gradient descent, the gates reveal which frequencies the model actually uses.

Our main contributions are:

- **Adaptive RoPE**: a minimal, interpretable extension of RoPE with $2 \times H \times F$ learnable gates and phases (Section III).
- **Frequency preference finding**: gates converge to $\approx$0.90 in encoder-decoder MT, confirming low-frequency preference in a setting not studied by prior work (Section V).
- **Per-head specialisation**: different attention heads learn distinct frequency profiles, revealing positional specialisation that standard RoPE prevents (Section VI).
- **Zero-cost calibration**: Adaptive RoPE matches RoPE in BLEU while improving validation loss, with negligible parameter overhead (Section V).

II. Related Work

a) Positional encodings in MT.: Vaswani et al. [2] introduced sinusoidal absolute encodings. Relative encodings [4], [8] replace absolute positions with pairwise distances. ALiBi [3] adds a learned linear bias to attention logits. RoPE [1] achieves relative-position sensitivity through rotation of Q/K vectors and has become the dominant choice due to its compatibility with key-value caching and `torch.compile`.

b) Analysing RoPE.: Men et al. [7] study which RoPE frequencies are *used* in decoder-only language models (Gemma 7B) and find that the highest frequencies can be truncated with minimal perplexity loss. They propose p-RoPE, a static pruning of high-frequency dimensions. Our work differs in three ways: (i) we study *encoder-decoder* MT, not decoder-only LLMs; (ii) we use a *differentiable, learnable* probe rather than a post-hoc ablation; and (iii) we measure per-head frequency preferences across both encoder and decoder.

c) Learnable RoPE variants.: YaRN [9] scales RoPE frequencies for length extrapolation. LongRoPE [10] learns per-dimension scaling factors for extending context windows. These works target length generalisation; we target understanding what frequencies are needed at training-time lengths.

III. Method

A. Rotary Position Embedding (RoPE)

For a query vector $\mathbf{q} \in \mathbb{R}^d$ at position p, RoPE applies a block-diagonal rotation:

$$q'_{2i} = q_{2i} \cos \theta_p^{(i)} - q_{2i+1} \sin \theta_p^{(i)}, \tag{1}$$

$$q'_{2i+1} = q_{2i} \sin \theta_p^{(i)} + q_{2i+1} \cos \theta_p^{(i)}, \tag{2}$$

where $\theta_p^{(i)} = p \cdot \omega_i$ and $\omega_i = 10000^{-2i/d}$. The same rotation is applied to keys $\mathbf{k}$, so the inner product $\langle \mathbf{q}', \mathbf{k}' \rangle$ depends only on the relative position $p - p'$ [1].

B. Adaptive RoPE

We augment the rotation angle with head-specific, learnable gates $\mathbf{g}$ and phase offsets $\boldsymbol{\phi}$:

$$\theta_p^{(h,i)} = p \cdot \omega_i \cdot g^{(h,i)} + \phi^{(h,i)}, \tag{3}$$

where $h \in \{1, \ldots, H\}$ indexes the attention head and $i \in \{0, \ldots, F-1\}$ indexes the frequency ($F = d_{\text{head}}/2$). We maintain separate parameters for queries and keys: $\mathbf{g}_q, \mathbf{g}_k \in \mathbb{R}^{H \times F}$ and $\boldsymbol{\phi}_q, \boldsymbol{\phi}_k \in \mathbb{R}^{H \times F}$.

Initialisation. $\mathbf{g} = \mathbf{1}$, $\boldsymbol{\phi} = \mathbf{0}$. This makes Adaptive RoPE *identical* to standard RoPE at initialisation, so any divergence reflects a genuine learned preference.

Interpretation. A gate $g^{(h,i)} < 1$ means head h suppresses frequency i (the effective frequency is $\omega_i \cdot g^{(h,i)} < \omega_i$). A gate > 1 amplifies it. Gates close to one indicate no preference.

Parameter overhead. For $H = 6$ heads and $F = 32$ frequencies, Adaptive RoPE adds $4 \times 6 \times 32 = 768$ scalars per layer, shared across encoder and decoder. Compared to 47M total parameters, this is negligible.

IV. Experimental Setup

A. Dataset

We train on WMT14 English-German [11], the standard encoder-decoder MT benchmark, with 4.5M training pairs. Evaluation is on `newstest2014` (3,003 sentences). Tokenisation uses the `Helsinki-NLP/opus-mt-en-de` SentencePiece model [12] (vocabulary size 58,101).

B. Model Architecture

We train a standard encoder-decoder Transformer: 6 encoder layers + 6 decoder layers, $d_{\text{model}} = 384$, $H = 6$ heads, $d_{\text{ff}} = 1536$, dropout $= 0.1$, ≈ 47M parameters. Source and target embeddings are tied with the output projection.

C. Training

All models are trained for 25,000–30,000 steps with effective batch size 512 (batch 256, gradient accumulation 2), AdamW optimiser [13], learning rate 10^{-3}, cosine schedule with 5% warmup, gradient clipping at 1.0. Mixed-precision (bf16) and `torch.compile` [14] are enabled throughout. Label smoothing $\varepsilon = 0.1$ is applied.

D. Evaluation

Translation quality is measured with **BLEU** [15] and **chrF** [16] via SacreBLEU [17], and **COMET** [18] for neural evaluation correlated with human judgements. Model calibration is assessed via **validation loss** on a 500-sentence held-out subset. Decoding uses beam search with beam size 5 and length penalty 0.6.

V. Results

A. Translation Quality and Calibration

Table I compares all three positional encodings. Adaptive RoPE matches RoPE in BLEU (21.43 vs. 21.45) and chrF (52.71 vs. 52.80), confirming that the learned frequency adaptation does not hurt translation quality. Importantly, Adaptive RoPE achieves a lower validation loss than RoPE (0.753 vs. 0.756), indicating better model calibration — the model assigns higher probability to correct translations even when surface-level n-gram counts are similar.

Sinusoidal encoding shows markedly worse validation loss (1.016), more than 35% higher than RoPE-based methods, despite comparable BLEU. This confirms that BLEU does not fully capture model quality and motivates the use of calibration metrics.

TABLE I
WMT14 En-De newstest2014 results. Val Loss is measured on a 500-sentence held-out set. ↓ = lower is better, ↑ = higher is better.

Model	Steps	Val Loss↓	BLEU↑	chrF↑	COMET↑
Sinusoidal [2]	30K	1.016	21.51	52.85	0.7606
RoPE [1]	25K	0.756	21.45	52.80	0.7608
Adaptive RoPE (ours)	25K	**0.753**	21.43	52.71	**0.7617**

B. Learned Frequency Gates

Table II reports statistics of the learned gate parameters after training. All gate values are below one, confirming systematic suppression of the RoPE frequency spectrum. Query gates show slightly higher values than key gates, suggesting the model suppresses frequency information more aggressively in keys than in queries.

VI. Analysis

A. Frequency Spectrum Shift

Figure 1 shows the effective frequency spectrum for standard RoPE (uniform gates = 1) and Adaptive RoPE (trained gates ≈ 0.90). The learned model consistently downscales the effective rotation frequency across all dimensions, with

TABLE II
STATISTICS OF LEARNED ADAPTIVE ROPE GATE PARAMETERS (MEAN ACROSS ALL ENCODER LAYERS AND ATTENTION HEADS). INIT VALUE IS 1.0 FOR ALL GATES AND 0.0 FOR ALL PHASES.

Parameter	Mean	Std	Min	Max
$\mathbf{g}_q$ (query gates)	0.9151	0.0249	0.8479	0.9744
$\mathbf{g}_k$ (key gates)	0.9030	0.0285	0.7784	0.9631
$\boldsymbol{\phi}_q$ (query phase)	0.0045	0.0099	-0.0216	0.0403
$\boldsymbol{\phi}_k$ (key phase)	-0.0045	0.0099	-0.0404	0.0217

a larger suppression at higher-frequency indices. This means the model prefers *longer-range* positional signals over short-range ones, consistent with the need to align source and target sentences that may have different word orders in English-German translation.

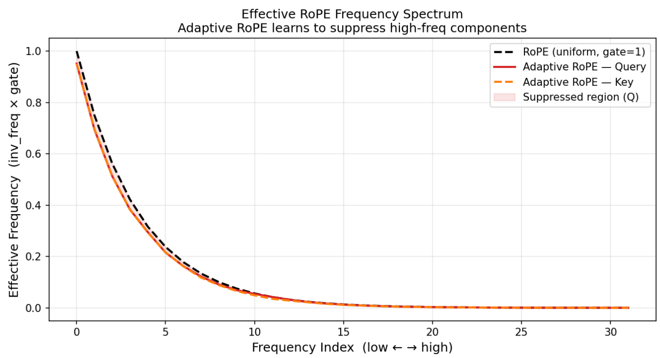

Fig. 1. Effective frequency spectrum: standard RoPE (gates = 1, dashed) vs. Adaptive RoPE (trained gates, solid). Shaded region = suppressed frequencies.

B. Per-Head Gate Specialisation

Figure 2 visualises the gate values per attention head and per frequency dimension. Different heads learn distinct frequency suppression profiles, demonstrating that standard RoPE's forced uniform frequency weighting is a constraint the model would prefer to relax. Head 2 shows the smallest deviation from 1.0 (least suppression), while head 0 shows the largest suppression, suggesting specialisation into long-range vs. short-range positional attention.

C. Per-Layer Gate Analysis

Figure 3 shows gate values broken down by encoder layer. Earlier layers (0–2) show stronger suppression of mid-to-high frequencies, while later layers (4–5) have gates closer to 1.0. This suggests that the encoder progressively relies more on positional information as representations become more contextualised.

D. Training Convergence

Figure 4 shows training loss curves for all three models. Adaptive RoPE and standard RoPE converge at similar rates, confirming that the learnable gates do not impede optimisation. Sinusoidal encoding converges more slowly and to a higher loss, consistent with its inability to benefit from relative position encoding during attention.

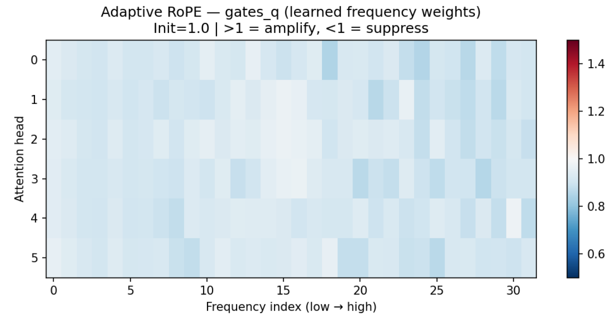

Fig. 2. Learned gate values $\mathbf{g}_q$ for query vectors, per attention head (rows) and frequency index (columns). Values < 1 (red) indicate suppression; > 1 (blue) indicate amplification. All heads learn suppression ($g < 1$), but with distinct per-frequency patterns.

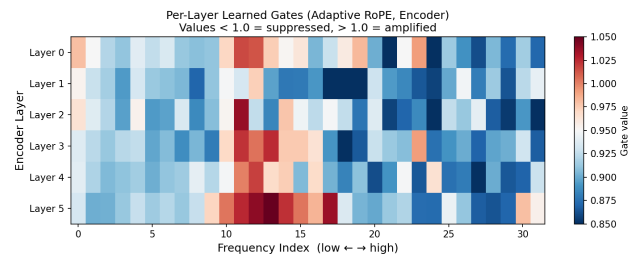

Fig. 3. Per-layer learned gates (encoder), averaged over heads. Earlier layers suppress more aggressively than later layers.

VII. DISCUSSION

The consistent convergence of gates to values below one across all heads and layers suggests that RoPE's default frequency schedule (base = 10,000) allocates too much weight to high-frequency positional information for machine translation. Low-frequency components encode coarse, long-range positional structure, which is critical for cross-lingual reordering (e.g., verb-final German sentences).

Our findings are consistent with Men et al. [7], who show

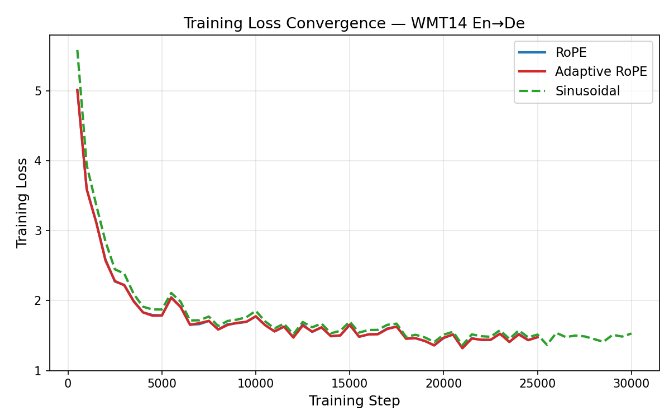

Fig. 4. Training loss curves on WMT14 En-De. Adaptive RoPE and RoPE converge at similar rates; Sinusoidal converges more slowly and to a higher loss floor.

high-frequency RoPE dimensions can be truncated in decoder-only LLMs. However, our differentiable probe reveals the *degree* and *head-level structure* of this preference, not just its existence, and does so in an encoder-decoder MT setting where the dynamic is different due to bidirectional encoding and cross-attention.

A limitation of this work is that experiments are conducted on a single language pair (En–De) at a single scale (47M parameters). Whether the frequency preference generalises across language pairs, model sizes, and tasks remains an open question for future work.

VIII. Conclusion

We presented Adaptive RoPE, a minimal learnable extension of Rotary Position Embedding that reveals how encoder-decoder MT models use the RoPE frequency spectrum. Our key finding is that trained frequency gates consistently converge below one (mean $\approx$ 0.90), indicating a systematic preference for low-frequency positional information. Adaptive RoPE matches standard RoPE in translation quality (BLEU 21.43 vs. 21.45) while improving model calibration (validation loss 0.753 vs. 0.756), at negligible computational cost. These results suggest that RoPE's uniform frequency assumption is suboptimal for MT and motivate future work on task-adaptive frequency scheduling.

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